**Evaluate Yourself:**

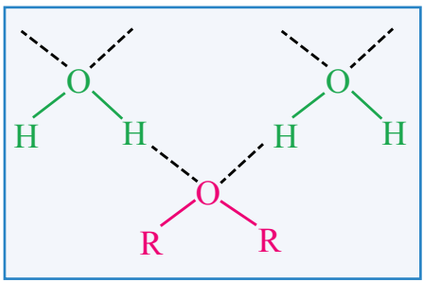
1. Which of the following reaction will give 1-methoxy-4-nitrobenzene.
   a) 4-nitro-1-bromobenzene + sodium methoxide.
   b) 4-nitrosodium phenoxide+bromomethane
2. Arrange the following compounds in the increasing order of their acid strength. propan-1-ol, 2,4,6-trinitrophenol, 3-nitrophenol, 3,5-dinitrophenol, phenol, 4-methylphenol.

## Physical Properties:

Ethers are polar in nature. The dipolemoment of ether is the vector sum of two polar C-O bonds with significant contribution from two lone pairs of electrons. For example, the dipole moment of diethyl ether is 1.18D. Boiling point of ethers are slightly higher than that of alkanes and lower than that of alcohols of comparable masses.

| Compound | Molar Mass | Boiling point |
|---|---|---|
| $CH_3-(CH_2)_5-CH_3$ n-heptane | 100.21 | 371K |
| $CH_3-O-(CH_2)_4-CH_3$ 1-methoxypentane | 102.17 | 373K |
| $CH_3-(CH_2)_5-OH$ hexan-1-ol | 102.16 | 430K |

Oxygen of ether can also form Hydrogen bond with water and hence they are miscible with water. Ethers dissolve wide range of polar and non-polar substances.




## Chemical Properties of ethers:

### 1. Nucleophilic substitution reactions of ethers.

Ethers can undergo nucleophilic substitution reactions with HBr or HI . HI is more reactive than HBr .

$$CH_3\text{-}O\text{-}CH_2\text{-}CH_3 + HI \xrightarrow{\Delta} CH_3I + CH_3\text{-}CH_2\text{-}OH$$

methoxy ethane — iodo methane — ethanol

$$C_6H_5\text{-}O\text{-}CH_3 + HI \longrightarrow C_6H_5\text{-}OH + CH_3I$$

methoxy benzene — Phenol — iodomethane



XII U11-Hydroxy compounds.indd 136 16-12-2022 19:42:20